TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Ultra-Precision Hybrid Machining Technology for SiC-Based CMC and EBC/TBC-Coated Gas Turbine Vanes and Blades**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Precision Manufacturing Machinery	*Cutting Machine Tools*		
Concept of Proposed Technology	**Core Concept** · **Target Components:** SiC-based CMC gas turbine vanes and blades with EBC/TBC coatings (without cooling holes) · **Key Technical Challenges:** Due to the anisotropic fiber architecture and brittle matrix/coating interfaces, machining often induces chipping, fiber pull-out, and subsurface damage. As a result, achieving ultra-low surface roughness and strict dimensional tolerances using conventional grinding alone is extremely difficult. · **Solution Strategy:** Establish an integrated hybrid machining–metrology platform incorporating the			

	following elements: • Five-axis ultrasonic-assisted grinding and milling for critical functional surfaces (roots, platforms, sealing areas, etc.) • Micro-abrasive waterjet (Micro-AWJ) for controlled trimming, edge conditioning, and coating pre-treatment/repair processes • Ultrashort-pulse laser trimming for localized geometry correction and minimal thermal damage (excluding drilling applications) • In-process sensing and closed-loop control to maintain surface integrity and process capability (C_p/C_{pk}) · **Deliverables:** Process window maps, inspection and acceptance gates (pass/fail criteria), pilot manufacturing recipes, and demonstration components validated through Doosan qualification processes
Rationale for Joint R&D Project	**Technical and Industrial Limitations** Leading global companies systematically manage CMC machining and coating finishing as core manufacturing competencies and have established highly repeatable and qualification-ready production systems. In contrast, although domestic industries have secured basic capabilities in CMC material development and production, advanced componentization technologies based on ultra-precision machining remain insufficiently mature. In particular, CMC components for gas turbine hot sections operate under extreme conditions and therefore require exceptionally high standards of surface integrity, dimensional accuracy, and coating reliability. However, conventional machining approaches alone are not capable of consistently meeting these stringent requirements. Furthermore, process development based primarily on empirical trial-and-error increases development time and cost, while amplifying technical risks during qualification and certification processes. **Necessity of International Collaboration** Purdue University possesses world-class expertise in machining science, ultrashort-pulse laser–material interactions, and advanced microstructural and defect analysis technologies, including micro-CT and SEM. These capabilities enable mechanism-based process optimization beyond empirical approaches, significantly accelerating technology development. International collaboration with such a leading research institution provides critical support for overcoming domestic technological limitations and internalizing advanced machining and inspection technologies. In addition, close coordination with system integrators and end-users, who provide system requirements, technical drawings, acceptance criteria, and qualification pathways, enables early alignment with commercialization objectives. This integrated collaboration framework facilitates early risk mitigation, shortens the transition to practical deployment and mass production, and ultimately

	ensures the successful industrialization of the developed technologies. Furthermore, the realization of a fully integrated smart welding system requires continuous validation through repeated on-site testing and iterative process optimization based on real manufacturing data. However, domestic development environments are often limited in terms of specialized research manpower and access to advanced domain-specific methodologies in ultra-precision machining, advanced manufacturing processes, and intelligent process monitoring technologies. In particular, the development of high-accuracy machining, adaptive process control, automated process planning, and intelligent quality assurance systems requires deep expertise in advanced manufacturing science and data-driven optimization, which cannot be sufficiently secured through conventional engineering approaches alone. In this context, collaboration with overseas research institutions possessing internationally recognized capabilities in these fields is essential. Such partnerships enable the transfer of advanced methodologies, reduce technological risks, and accelerate process stabilization and system optimization. Purdue University possesses world-class research capabilities in metal forming process physics modeling, digital twin–based manufacturing analysis, and physics-informed AI for manufacturing processes. In addition, Purdue operates specialized research centers such as the Manufacturing and Materials Research Laboratory (MMRL) and the Composite Materials Simulation Center (CMSC). These centers provide advanced research infrastructure and interdisciplinary expertise in precision machining, composite material modeling, process simulation, and data-driven manufacturing system development, serving as critical platforms for integrating fundamental research with practical industrial implementation. As a manufacturing-focused engineering university located in Indiana, one of the major industrial hubs in the United States, Purdue maintains strong partnerships with medium- and large-scale manufacturing enterprises and actively utilizes large-scale equipment and production data for smart manufacturing and advanced process research. The university also operates testbed-based research environments and benefits from its strategic location within a major transportation network. This enables direct observation of technological trends and market demands in the U.S. manufacturing sector. Consequently, collaboration with Purdue University provides a unique opportunity to establish a comprehensive development framework that links industrial needs with core precision manufacturing technologies, field validation, process refinement, and commercialization, thereby enhancing technological maturity, field applicability, and global competitiveness.
Key Words	**1. CM) Machining** **2. SiC/SiC and C/SiC Composite Materials** **3. Rotary Ultrasonic Machining (RUM)** **4. Micro-Abrasive Waterjet (Micro-AWJ) Processing** **5. EBC/TBC Surface Finishing** **6. Surface Integrity Control** **7. Closed-Loop Process Control**

(max 10 pages)

	8. **Gas Turbine Vanes and Blades**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	**R&D Objectives** The objective of this project is to establish a world-class, highly repeatable manufacturing process window for SiC-based CMC gas turbine vanes and blades with EBC/TBC coatings, enabling stable, high-quality, and qualification-ready production. **Detailed Objectives** This project aims to develop a highly reliable and repeatable hybrid machining and finishing platform for SiC-based ceramic matrix composite (CMC) gas turbine vanes and blades with environmental and thermal barrier coatings (EBC/TBC). The system will integrate five-axis ultrasonic-assisted grinding and milling, micro-abrasive waterjet polishing, and ultrashort-pulse laser trimming to enable precise and damage-minimized material removal across complex turbine geometries. To ensure superior surface integrity, the project will implement in-process sensing and closed-loop control technologies to actively monitor and regulate machining conditions. This approach will minimize surface chipping, fiber pull-out, subsurface damage, and coating delamination, thereby preserving both structural and functional performance. Advanced surface integrity management will be a core element in establishing a stable and scalable process window. In addition, the project will develop qualification-oriented metrology and inspection frameworks that integrate three-dimensional geometry measurement, surface roughness evaluation, subsurface damage assessment, and coating integrity verification. Standardized acceptance criteria and pass/fail gates will be established to support certification, quality assurance, and industrial deployment. Finally, the developed technologies will be validated through demonstration on representative CMC gas turbine vanes or blades without cooling holes, including EBC/TBC coatings. The project will achieve target functional surface quality on critical aerodynamic and thermal protection areas, demonstrating the feasibility of the proposed process for high-performance turbine component manufacturing and commercialization.			
R&D Target Specifications	[A-Grade: Functional Surfaces (Platform, Root/Dovetail, Sealing/Contact Areas)] · Surface Roughness: Ra ≤ 0.20 μm (target), Rz ≤ 1.5 μm · Edge Quality: Maximum chipping length ≤ 50 μm; chipping frequency ≤ 3 per 100 mm · Subsurface Damage Depth: ≤ 10 μm (based on CT/cross-sectional correlation protocol) · Geometric Tolerances: Profile/flatness/cylindricity ≤ 20 μm (part-dependent), repeatability Cpk ≥			

(max 10 pages)

	1.33 [B-Grade: General Aerodynamic Surfaces] · Surface Roughness: Ra ≤ 0.8 μm; geometric tolerance ≤ 50 μm [Coating (EBC/TBC) Surfaces] · After Coating Finishing: Ra ≤ 0.50 μm (topcoat surface) · Coating Thickness Uniformity: ±10 μm (or ±10% for highly curved geometries) · No delamination or through-thickness cracking after thermal cycling screening (project-defined test matrix) [Manufacturability Metrics] · Pilot Run FPY (First Pass Yield) ≥ 95% · Cycle Time Reduction ≥ 30% compared to baseline (grinding-only process)
R&D Plan	**Year 1 (TRL 3 → 4):** · Define a defect taxonomy and acceptance gates (chipping, fiber pull-out, subsurface damage, coating integrity) · Establish baseline process routes for CMC grinding and coating finishing, and develop metrology correlation frameworks (3D scanning, surface roughness, CT, cross-sectional analysis) · Develop process models and design of experiments (DoE) for ultrasonic-assisted machining and coating finishing processes **Year 2 (TRL 4 → 5):** · Integrate a hybrid machining cell: ultrasonic-assisted machining + micro-AWJ trimming/conditioning + (if required) ultrashort-pulse laser trimming · Implement in-process monitoring (spindle power, acoustic emission, vibration, temperature) and adaptive control logic · Validate the processes using representative vane/blade coupons and sub-components (platform, fillet, root geometries) **Year 3 (TRL 5 → 6):** · Demonstrate repeatable machining on full-scale CMC vanes/blades with coatings and achieve A/B-grade KPIs · Conduct pilot manufacturing runs, evaluate process capability (Cp/Cpk), and draft manufacturing specifications and inspection standard packages · Establish follow-on commercialization plans, including scale-up, cost reduction, and certification/qualification pathways in collaboration with Doosan

4. R&R (Roles and Responsibilities)

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Expected R&R of Research Institute	· Leading fundamental and applied research on CMC/coating surface integrity mechanisms, including subsurface damage, residual stress, and delamination · Developing machining physics-based models and DoE frameworks, and supporting ultrashort-pulse laser and acoustic emission (AE)-based monitoring algorithms · Providing advanced analytical capabilities, including micro-CT and SEM analysis, quantitative subsurface damage assessment, and coating microstructure evaluation · Supporting joint team training, academic dissemination of non-sensitive results, and international standardization activities when necessary
Expected R&R of Korean SME Partner	· Manufacturing of CMC composites and fabrication of prototype components · Establishing and operating a hybrid machining cell (ultrasonic-assisted grinding/milling, micro-AWJ, finishing grinding, and polishing) · Developing tools, fixturing, and process recipes for vane/blade geometries, acquiring process data, and implementing closed-loop control · Fabricating CMC demonstration components, conducting pilot production runs, and documenting manufacturing specifications, control plans, and PFMEA · Coordinating with end-user requirements, interpreting technical drawings, and supporting acceptance testing and logistics operations

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** This project will establish ultra-precision machining capabilities achieving surface roughness levels of approximately Ra 0.20 μm on functional surfaces of SiC-based CMC gas turbine vanes and blades, accompanied by systematic quantification and control of surface integrity. Through advanced hybrid machining and finishing technologies, the project will enable stable suppression of chipping, fiber pull-out, subsurface damage, and coating delamination, ensuring high structural reliability and long-term durability of critical turbine components. In addition, the project will develop standardized finishing and repair preparation process routes for EBC/TBC coatings, enabling precise control of surface roughness and coating thickness uniformity. These process routes will support both initial manufacturing and refurbishment operations, thereby extending component service life and reducing lifecycle costs. Furthermore, an integrated standard technology package will be established, incorporating defect classification systems, acceptance criteria, process window maps, and closed-loop machining control prototypes. This package will serve as a practical guideline for qualification, quality assurance, and industrial deployment. **Industrial Impact** The successful implementation of this project will support the establishment of a strategic domestic

	supply chain for next-generation gas turbine hot-section components, based on qualification and approval by major turbine system manufacturers. By meeting stringent aerospace-grade manufacturing and certification requirements, domestic companies will be positioned as reliable long-term suppliers for high-performance turbine components. Moreover, the project will significantly reduce dependence on overseas machining and repair technologies by internalizing core manufacturing know-how and establishing scalable production platforms. This will enable efficient expansion toward higher technology readiness levels (TRL 7–8), facilitate cost reduction and volume production, and strengthen the competitiveness of domestic manufacturing ecosystems. Finally, the project will enhance bilateral R&D collaboration between Korea and the United States and promote the establishment of advanced aerospace-grade manufacturing systems. Through sustained technology exchange, joint validation, and harmonization with international standards, the project will contribute to strengthening global competitiveness, expanding export opportunities, and positioning domestic industries as leaders in high-value-added turbine component manufacturing.

6. References

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